

RESEARCH EXPERIENCE

I build stable and trustworthy multimodal intelligence that unifies perception and generation across domains, modalities, and real-world settings. My research spans open-world and 3D perception, video/world generation, and AI for science, with the long-term goal of enabling general-purpose AI for digital life, autonomy, healthcare, and scientific discovery.

- AI for Digital Twins 2025 - Present
 - World Model; Digital Human
- AI for Autonomous Mobility 2020 - Present
 - 3D Occupancy Prediction; Open-vocabulary/Cross-domain/Open-set Object Detection
- AI for Scientific Innovation 2024 - 2025
 - Nanophotonics; Meta Optics; Graph-based Learning
- AI for Transforming Medicine 2020 - 2024
 - MRI Brain Analysis; Medical Report Generation; 4D Surgical Simulation

PROFILE HIGHLIGHTS

- Academic & Industrial Perspective
 - Founded a seed-funded Hong Kong startup while pursuing top-tier research (50+ papers; 15+ lead-author papers).
- Large-Scale GPU Experimentation
 - Extensive hands-on experience with large-scale GPU clusters (10,000+ GH200).
- Leadership & Collaboration
 - Led a Swiss national AI initiative with 15 leading PIs; NeurIPS workshop organizer.
- Rapid Citation Impact
 - U-KAN (co-first author) received 600+ citations within 1.5 years.
- Impactful First-Author Project
 - One first-author project attracted 2.4k+ GitHub stars and was deployed on the Poe platform.

RESEARCH HIGHLIGHTS

- Selected as Oral/Spotlight/Highlight Presentation (1st Author)
 - AAAI 2022; CVPR 2022; ICCV 2023; ECCV 2024; ICCV 2025; NeurIPS 2025; ICLR 2026
- Selected as Best Paper Finalist (1st Author)
 - CVPR 2022 (33/8161)
- Full-Score Submission (1st Author)
 - ICCV 2025, final scores: 6, 6, 6
- Selected as Cover Paper (2nd Author)
 - ACS Photonics, 2024
- AAAI 2025 Top-1 Influential Paper (Co-1st Author)
 - U-KAN: medical image segmentation and generation backbone

WORK EXPERIENCE

- École Polytechnique Fédérale de Lausanne (EPFL) Feb. 2025 - Present
 - Research Scientist, Robotics and Computer Vision
 - Supervisor: Prof. Alexandre Alahi (Advised by Prof. Feifei Li@Stanford)
- The Chinese University of Hong Kong (CUHK) Mar. 2024 - Feb. 2025
 - Postdoc, Electronic Engineering
 - Supervisor: Prof. Yixuan Yuan
- ScholaGO Education Technology Company Limited, Hong Kong 2024 - Present
 - Founder & CEO
 - Developed multimodal educational products; secured seed funding from HK Tech 300 and HKSTP.

EDUCATION

- City University of Hong Kong (CityU) Sep. 2020 - Feb. 2024
 - Ph.D., Electrical Engineering
 - Supervisor: Prof. Yixuan Yuan
 - 3.5-Year; Early Graduation
- Tianjin University (TJU) Sep. 2016 - Jun. 2020
 - B.S., Electrical Engineering
 - GPA: 3.83/4.00 (Top 2%)

1. **[Digital Human] EverAnimate: Minute-Scale Human Animation via Latent Flow Restoration**
Wuyang Li, Yang Gao, Mariam Hassan, Lan Feng, Wentao Pan, Po-Chien Luan, Alexandre Alahi
 Under Review, 2026
2. **[World Model] SurgVista: Long-Horizon Surgical World Modeling with Plausible Instrument-Tissue Dynamics**
 Wentao Pan*, **Wuyang Li**^{†,*}, Shengyuan Liu, Xinyu Liu, Hengyu Liu, Yixuan Yuan
 Under Review, 2026
3. **[World Model] Drift-Resistant Navigation World Model with Anchored Epipolar Guidance**
 Po-Chien Luan, Zimin Xia, **Wuyang Li**, Yang Gao, Alexandre Alahi
 Under Review, 2026
4. **[Video Generation] Anchored Video Generation: Decoupling Scene Construction and Temporal Synthesis in Text-to-Video Diffusion Models**
 Mariam Hassan, Bastien Van Delft, **Wuyang Li**[†], Alexandre Alahi
 Under Review, 2026, High-score ECCV submission
5. **[Video Generation] Social-Mamba: Socially-Aware Trajectory Forecasting with State-Space Models**
 Po-Chien Luan, **Wuyang Li**[†], Yang Gao, Alexandre Alahi
 Under Review, 2026, High-score ECCV submission
6. **[Video Generation] Stable Video Infinity: Infinite-Length Video Generation with Error Recycling**
Wuyang Li, Wentao Pan, Po-Chien Luan, Yang Gao, Alexandre Alahi
 International Conference on Learning Representations (ICLR), 2026, **Oral (1.3%)**
7. **[3D Perception] Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation**
 Yang Gao, **Wuyang Li**[†], Po-Chien Luan, Alexandre Alahi
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026
8. **[3D Perception] VoxDet: Rethinking 3D Semantic Occupancy Prediction as Dense Object Detection**
Wuyang Li, Zhu Yu, Alexandre Alahi
 Neural Information Processing Systems (NeurIPS), 2025, **Spotlight (3%)**
9. **[AI for Science] MetaScope: Optics-Driven Neural Network for Ultra-Micro Metalens Endoscopy**
Wuyang Li^{*}, Wentao Pan*, Xiaoyuan Liu*, Zhendong Luo, Chenxin Li, Hengyu Liu, Din Ping Tsai, Mu Ku Chen, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2025, **Full final scores (6, 6, 6)**
10. **[Backbone] U-KAN Makes a Strong Backbone for Medical Image Segmentation and Generation**
 Chenxin Li*, Xinyu Liu*, **Wuyang Li**^{*}, Cheng Wang*, Hengyu Liu, Yifan Liu, Zhen Chen, Yixuan Yuan
 Association for the Advancement of Artificial Intelligence (AAAI), 2025, **AAAI 2025 Top-1 Influential Paper**
11. **[2D Perception] CLIFF: Continual Latent Diffusion for Open-Vocabulary Object Detection**
Wuyang Li, Xinyu Liu, Jiayi Ma, Yixuan Yuan
 European Conference on Computer Vision (ECCV), 2024, **Oral (1.8%)**
12. **[Video Generation] Boosting Medical Image Analysis via Motion-Informed Generative Videos**
Wuyang Li, Xinyu Liu, Qiushi Yang, Yixuan Yuan
 Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
13. **[Open-World Learning] Novel Scenes & Classes: Towards Adaptive Open-set Object Detection**
Wuyang Li, Xiaoqing Guo, Yixuan Yuan
 IEEE International Conference on Computer Vision (ICCV), 2023, **Oral (1.8%)**
14. **[Open-World Learning] Adjustment and Alignment for Unbiased Open Set Domain Adaptation**
Wuyang Li, Jie Liu, Bo Han, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023
15. **[Domain Shift] SIGMA++: Improved Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023
16. **[Domain Shift] SIGMA: Semantic-complete Graph Matching for Domain Adaptive Object Detection**
Wuyang Li, Xinyu Liu, Yixuan Yuan
 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022, **Best Paper Finalist (0.4%)**

17. **[Domain Shift]** SCAN: Cross Domain Object Detection with Semantic Conditioned Adaptation
Wuyang Li, Xinyu Liu, Xiwen, Yao, Yixuan Yuan
Association for the Advancement of Artificial Intelligence (AAAI), 2022, **Oral (4.5%)**
18. **[Brain Analysis]** Synthesizing Realistic fMRI: A Physiological Dynamics-Driven Hierarchical Diffusion Model for Efficient fMRI Acquisition
Yufan Hu, Yu Jiang, Wuyang Li, Yixuan Yuan
International Conference on Learning Representations (ICLR), 2025
19. **[Brain Analysis]** InstantSplamp: Fast and Generalizable Stenography Framework for Generative Gaussian Splatting
Chenxin Li*, Hengyu Liu*, Zhiwen Fan, Wuyang Li, Yifan Liu, Panwang Pan, Yixuan Yuan
International Conference on Learning Representations (ICLR), 2025
20. **[Domain Shift]** Universal Domain Adaptive Object Detection via Dual Probabilistic Alignment
Yuanfan Zheng*, Jinlin Wu*, Wuyang Li, Zhen Chen
Association for the Advancement of Artificial Intelligence (AAAI), 2025
21. **[AI for Science]** Stereo Vision Meta-Lens-Assisted Driving Vision
Xiaoyuan Liu, Wuyang Li, Takeshi Yamaguchi, Zihan Geng, Takuo Tanaka, Din Ping Tsai, Mu Ku Chen
ACS Photonics, 2024, **Selected as Cover Paper**
22. **[SAM]** Flaws can be Applause: Unleashing Potential of Segmenting Ambiguous Objects in SAM
Chenxin Li*, Yuzhi Huang*, Wuyang Li, Hengyu Liu, Xinyu Liu, Qing Xu, Zhen Chen, Yue Huang, Yixuan Yuan
Conference on Neural Information Processing Systems (NeurIPS), 2024
23. **[SAM]** When 3D Partial Points Meets SAM: Tooth Point Cloud Segmentation with Sparse Labels
Yifan Liu, Wuyang Li, Cheng Wang, Hui Chen, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024, **Early Accept (14.0%)**
24. **[Diffusion]** Latent Diffusion Label Rectification for Semi-supervised Medical Image Segmentation
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
25. **[Brain]** fMRI to T1w MRI Knowledge Transfer Network for Brain Multi-phenotype Prediction
Zhibin He, Wuyang Li, Yu Jiang, Zhihao Peng, Pengyu Wang, Xiang Li, Tianming Liu, Junwei Han, et al.
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
26. **[3D]** LGS: A Light-weight 4D Gaussian Splatting for Efficient Surgical Scene Reconstruction
Hengyu Liu*, Yifan Liu*, Chenxin Li*, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
27. **[Video Generation]** Endora: Video Generation Models as Endoscopy Simulators
Chenxin Li, Hengyu Liu, Yifan Liu, Brandon Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing Shao, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024
28. **[Brain]** Foundation Model for Any Phenotype Prediction via fMRI to sMRI Knowledge Transfer
Zhibin He, Wuyang Li, Yifan Liu, Xinyu Liu, Junwei Han, Tuo Zhang, Yixuan Yuan
IEEE Transactions on Medical Imaging (TMI), 2024
29. **[Multi-Modality]** MRM: Masked Relation Modeling for Medical Image Pre-Training with Genetics
Qiushi Yang, Wuyang Li, Baopu Li, Yixuan Yuan
IEEE International Conference on Computer Vision (ICCV), 2023
30. **[Noisy Labels]** Intervention & Interaction Federated Abnormality Detection with Noisy Clients
Xinyu Liu, Wuyang Li, Yixuan Yuan
Medical Image Computing and Computer Assisted Intervention (MICCAI), 2022, **Oral, Early Accept**
31. **[Noisy Labels]** Towards Robust Adaptive Object Detection under Noisy Annotations
Xinyu Liu, Wuyang Li, Qiushi Yang, Baopu Li, Yixuan Yuan
IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022
32. **[Object Detection]** HTD: Heterogeneous Task Decoupling for Two-Stage Object Detection
Wuyang Li, Zhen Chen, Baopu Li, Dingwen Zhang, Yixuan Yuan
IEEE Transactions on Image Processing (TIP), 2021

AWARDS

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- **Outstanding Academic Performance Award, CityU** 2022 - 2023
Outstanding research achievements
 - **Research Tuition Scholarship, CityU** 2022 - 2023
Outstanding research achievements
 - **Ranked First in Polyp Detection Challenge** 2021
3rd International Endoscopy Computer Vision Challenge and Workshop (EndoCV2021)
 - **Undergraduate National Scholarship** 2017 - 2018
The highest undergraduate honor for a student in China
 - **Silver Award in Tianjin Advanced Mathematical Competition** 2017
As a selected participant representing Tianjin University

SERVICE

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- **Leader of Large Swiss AI Initiative Project [Link]** 2025
Led a national project with 15 leading Swiss PIs; gained large-scale GPU cluster experience (10,000+ GH200).
 - **NeurIPS 2024 Workshop Organizer [Link]** 2024
Organized the Advancements in Multimodal Medical Foundation Models workshop
 - **Research Funding Proposals** 2022 - 2024
Assisted my advisor in preparing several competitive research funding proposals
 - **Startup Funding [Link]** 2023
Secured seed funding from HK Tech 300 and HKSTP
 - **Founder, ScholaGO Education Technology Company Limited, Hong Kong [Link]** 2024
Founded a company developing multimodal educational products
 - **Conference and Journal Reviewer**
ICLR/ICML/NeurIPS; CVPR/ICCV; TPAMI; IJCV; TIP; NeurIPS **Top Reviewer**